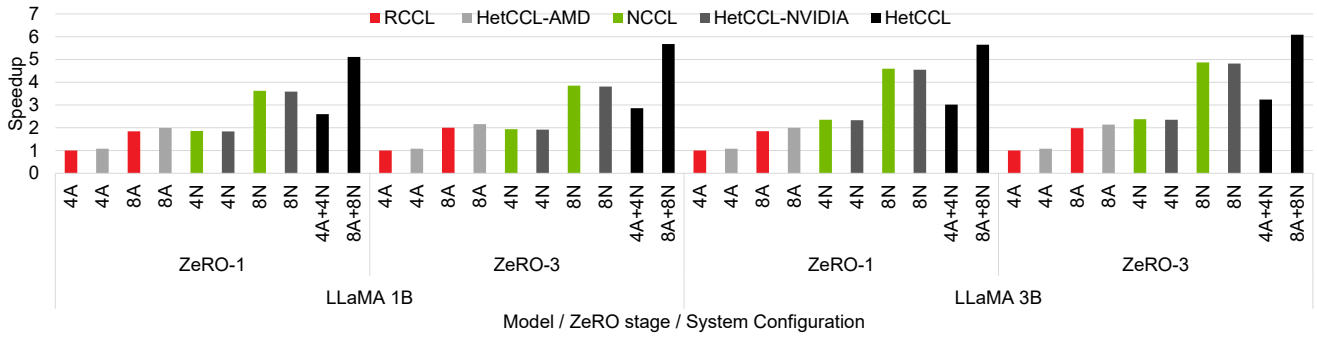
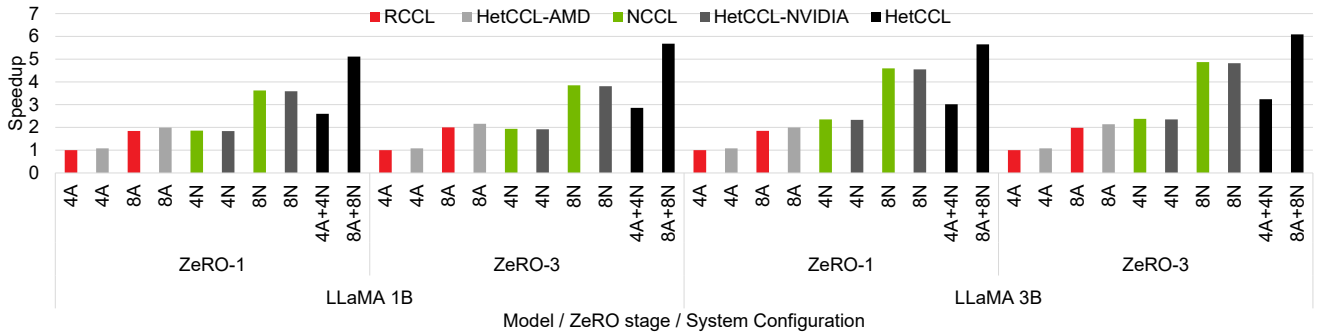
Table 1: End-to-end training latency breakdown across models.

Model	Communication (%)	Computation (%)	Synchronization (%)
GPT 125M	13.26	86.72	0.02
GPT 355M	16.96	83.03	0.01
LLaMA 1B	22.85	77.14	0.01
LLaMA 3B	29.09	70.91	0.00

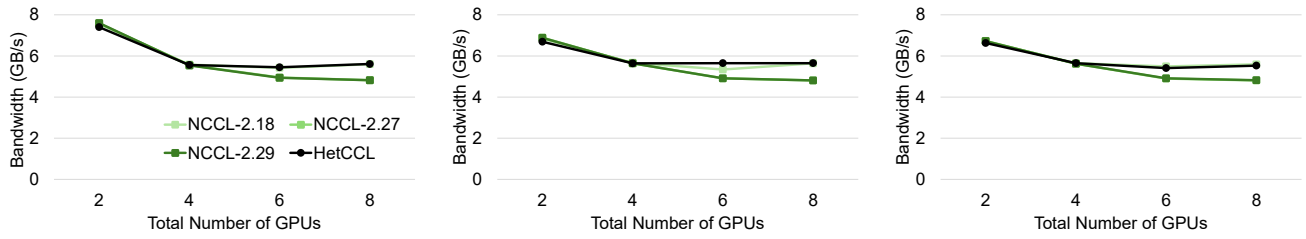


(a) GPT training



(b) LLaMA training

Figure 1: Speedup in LLM training. 4A/8A and 4N/8N represent training runs on 4/8 AMD GPUs and 4/8 NVIDIA GPUs, respectively. 4A+4N/8A+8N denotes a heterogeneous setting with 4/8 AMD and 4/8 NVIDIA GPUs.



(a) All-Reduce (b) All-Gather (c) Reduce-Scatter

Figure 2: Comparison of collective communication performance across NCCL versions 2.18, 2.27, and 2.29.

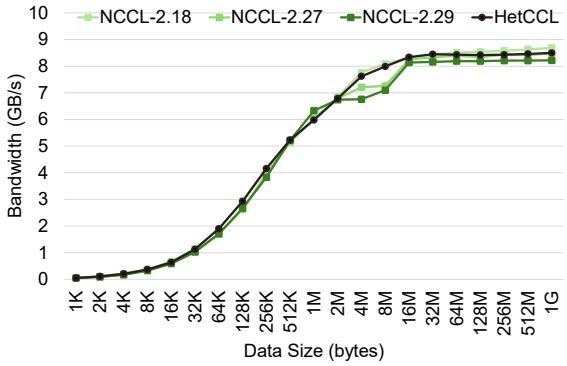


Figure 3: Comparison of point-to-point communication performance across NCCL versions 2.18, 2.27, and 2.29.

Table 2: HetCCL training efficiency normalized by the NCCL/RCCL baselines. Efficiency is defined as the throughput of HetCCL on 8 NVIDIA + 8 AMD GPUs divided by the sum of the throughput of NCCL on 8 NVIDIA GPUs and that of RCCL on 8 AMD GPUs.

Model	Efficiency
GPT 125M	0.91
GPT 355M	0.91
LLaMA-3 1B	0.97
LLaMA-3 3B	0.90
LLaMA-2 7B	0.91
LLaMA-2 13B	0.93

Table 3: LLaMA-2 7B and 13B model training with Megatron-DeepSpeed.

CCL (# of GPUs)	LLaMA-2 7B		LLaMA-2 13B	
	Throughput (tokens/s)	Speedup	Throughput (tokens/s)	Speedup
RCCL (8xAMD)	691	1.00	386	1.00
NCCL (8xNVIDIA)	1289	1.87	716	1.85
HetCCL (4xAMD + 4xNVIDIA)	932	1.35	537	1.39
HetCCL (8xAMD + 8xNVIDIA)	1806	2.61	1021	2.64

Table 4: Sparse MoE model training with expert parallelism.

CCL (# of GPUs)	Throughput (tokens/s)	Speedup
RCCL (8xAMD)	2792	1.00
NCCL (8xNVIDIA)	4287	1.54
HetCCL (4xAMD + 4xNVIDIA)	3435	1.23
HetCCL (8xAMD + 8xNVIDIA)	6441	2.31